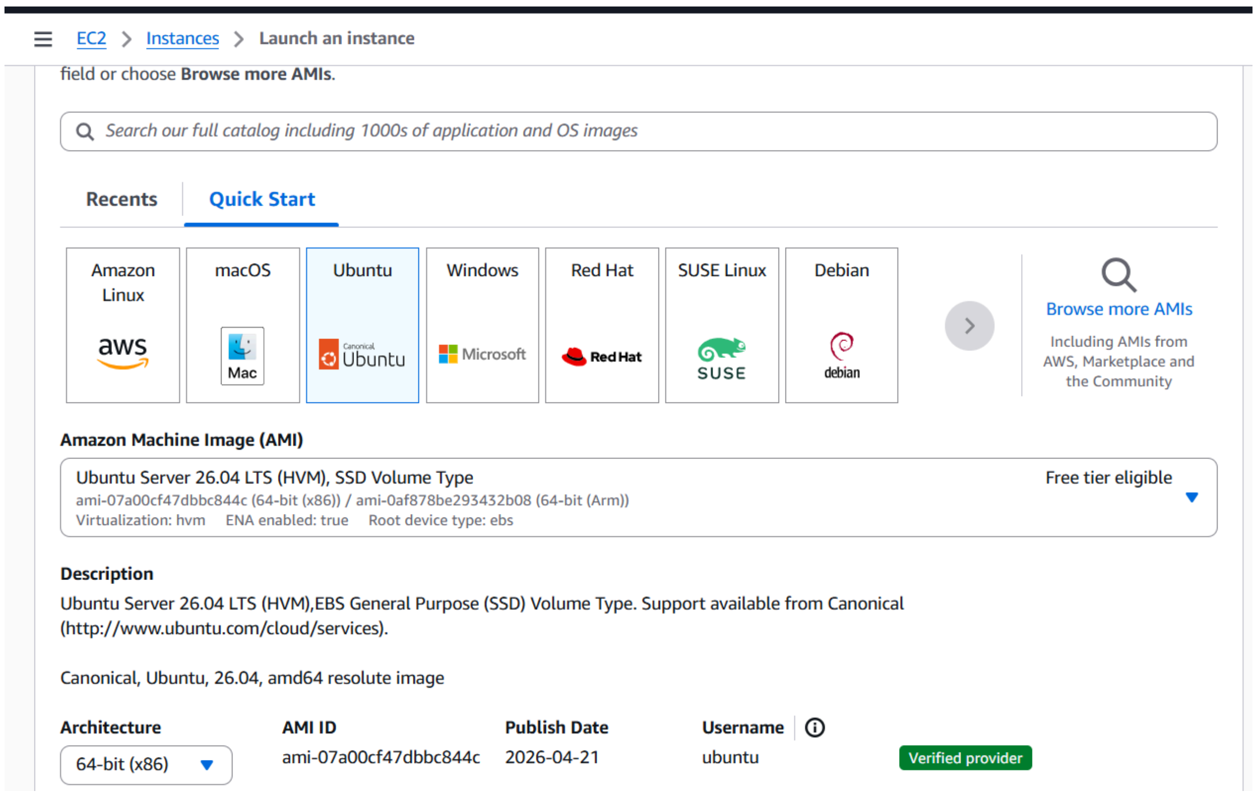
Docker-01

1. Setup a virtual machine using Ubuntu image.

Create an Ubuntu EC2 instance



2. Install docker on ec2.

Update packages: apt update

```
ubuntu@ip-172-31-10-98:~$ sudo apt update
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [136 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 Packages [1483 kB]
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main Translation-en [524 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64 Components [395 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64 c-n-f Metadata [32.4 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages [16.3 MB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security InRelease [136 kB]
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:11 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 Components [4556 kB]
```

Install Docker: sudo apt install docker.io -y

```
ubuntu@ip-172-31-10-98:~$ sudo apt install docker.io -y
Installing:
  docker.io

Installing dependencies:
  bridge-utils  containerd  dns-root-data  dnsmasq-base  pigz  runc  ubuntu-fan

Suggested packages:
  ifupdown  aufs-tools  cgroupfs-mount  | cgroup-lite  debootstrap  docker-buildx  docker-compose-v2

Summary:
  Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 39
  Download size: 74.0 MB
  Space needed: 281 MB / 48.4 GB available
```

Enable and start Docker:

```
ubuntu@ip-172-31-10-98:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-05-30 15:20:10 UTC; 3min 5s ago
 Invocation: 0b63d9e2468643d385fb64a55388fdfb
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 2334 (dockerd)
      Tasks: 9
     Memory: 25.2M (peak: 25.5M)
        CPU: 220ms
     CGroup: /system.slice/docker.service
             └─2334 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

May 30 15:20:10 ip-172-31-10-98 dockerd[2334]: time="2026-05-30T15:20:10.746247855Z" level=info msg="Restoring containers: start
May 30 15:20:10 ip-172-31-10-98 dockerd[2334]: time="2026-05-30T15:20:10.769193048Z" level=info msg="Deleting nftables IPv4 rule
May 30 15:20:10 ip-172-31-10-98 dockerd[2334]: time="2026-05-30T15:20:10.773687537Z" level=info msg="Deleting nftables IPv6 rule
May 30 15:20:10 ip-172-31-10-98 dockerd[2334]: time="2026-05-30T15:20:10.961873605Z" level=info msg="Loading containers: done."
```

Check Docker version:

```
ubuntu@ip-172-31-10-98:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
ubuntu@ip-172-31-10-98:~$
```

```
ubuntu@ip-172-31-10-98:~$ sudo usermod -aG docker ubuntu
ubuntu@ip-172-31-10-98:~$ newgrp docker
ubuntu@ip-172-31-10-98:~$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:0e760fdfbc48ba8041e7c6db999bb40bfca508b4be580ac75d32c4e29d202ce1
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
```

3. Pull 5 docker images. (nginx, Apache tomcat, ubuntu, Jenkins, SonarQube)

Pull images one by one:

Nginx :docker pull nginx

```
ubuntu@ip-172-31-10-98:~$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
7f8b1a2b17d8: Pull complete
45381ecb0e2f: Pull complete
b4a248c845e5: Pull complete
5b4d6ff92fc4: Pull complete
5431d0092ffd: Pull complete
830625e1ac85: Pull complete
13fd728be9eb: Pull complete
5e6b66b5e5f1: Download complete
cc5f57206478: Download complete
Digest: sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
ubuntu@ip-172-31-10-98:~$
```

Apache Tomcat : docker pull tomcat

```
ubuntu@ip-172-31-10-98:~$ docker pull tomcat
Using default tag: latest
latest: Pulling from library/tomcat
4f4fb700ef54: Pull complete
b40150c1c271: Pull complete
d92a694d8aa0: Pull complete
97c47a5c1a2f: Pull complete
b38893372039: Pull complete
6b82a07ff6e7: Pull complete
65313903a8a8: Pull complete
f80ef884f8c2: Download complete
13883d0c1f97: Download complete
Digest: sha256:247650787b97751d68aa2d532d6e09111c36946d9a50ea0e5071e19facc1e5e7
Status: Downloaded newer image for tomcat:latest
docker.io/library/tomcat:latest
ubuntu@ip-172-31-10-98:~$
```

Ubuntu: docker pull ubuntu

```
ubuntu@ip-172-31-10-98:~$ docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
1c24335ddd46: Pull complete
6f5c5aa4e145: Pull complete
9bcf140d7f0f: Download complete
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
ubuntu@ip-172-31-10-98:~$
```

Jenkins: docker pull jenkins

```
ubuntu@ip-172-31-10-98:~$ docker pull jenkins/jenkins:lts
lts: Pulling from jenkins/jenkins
cebc1e77ceba: Pull complete
3bdbd616c406: Pull complete
6b35bb76b8c8: Pull complete
801814f62c99: Pull complete
c2314d42cca1: Pull complete
76a355a69c2e: Pull complete
5834ae4b8774: Pull complete
ae218d1cfa6e: Pull complete
9014a3401c36: Pull complete
4c140080734e: Pull complete
dd00b4ea1f64: Pull complete
307f8152a55e: Pull complete
653a4e57bccf: Download complete
Digest: sha256:01c992ffef29dcf41c7164e8c16285c657c2368c4943dda8d68c93fdf54447d5
Status: Downloaded newer image for jenkins/jenkins:lts
docker.io/jenkins/jenkins:lts
ubuntu@ip-172-31-10-98:~$
```

Sonarqube : docker pull sonarqube

```
ubuntu@ip-172-31-10-98:~$ docker pull sonarqube
Using default tag: latest
latest: Pulling from library/sonarqube
4f4fb700ef54: Pull complete
c10b8232386e: Pull complete
f4306f81d85a: Pull complete
907ea4a8bc72: Download complete
8506693bc963: Download complete
Digest: sha256:35bedac3f40cda75969890da59b17d577770844fe6ef659206c678a8e00921c7
Status: Downloaded newer image for sonarqube:latest
docker.io/library/sonarqube:latest
ubuntu@ip-172-31-10-98:~$
```

Verify images: docker images

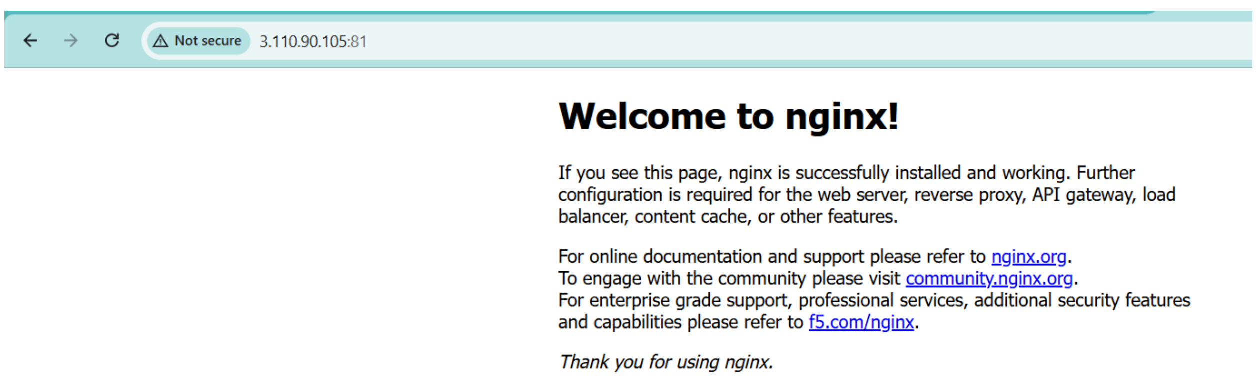
```
ubuntu@ip-172-31-10-98:~$ docker images

IMAGE                  ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest     0e760fdfbc48       25.9kB         9.49kB   U
jenkins/jenkins:lts    01c992ffef29        799MB          289MB
nginx:latest           5aca99593157        241MB           66MB
sonarqube:latest       35bedac3f40c       2.52GB         1.08GB
tomcat:latest          247650787b97        589MB          158MB
ubuntu:latest          f3d28607ddd7        160MB         45.3MB
ubuntu@ip-172-31-10-98:~$
```

4. Run nginx container and expose on port 81.

Run container: docker run -d --name nginx-container -p 81:80 nginx
Check running containers: docker ps

```
ubuntu@ip-172-31-10-98:~$ docker run -d --name nginx-container -p 81:80 nginx
9dddb9bd4569c5ab8fa7995b2c6e2ef0df1bde03f3f70664c2fbca539bca2642
ubuntu@ip-172-31-10-98:~$ 
```

Access in browser:



5. Delete the Apache tomcat image from local.

Delete Tomcat image: docker rmi tomcat
docker rm -f <container-id>

docker rmi tomcat

```
ubuntu@ip-172-31-10-98:~$ docker images

IMAGE                  ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest     0e760fdfbc48       25.9kB         9.49kB    U
jenkins/jenkins:lts    01c992ffef29        799MB          289MB
nginx:latest           5aca99593157        241MB           66MB    U
sonarqube:latest       35bedac3f40c       2.52GB         1.08GB
tomcat:latest          247650787b97        589MB          158MB
ubuntu:latest          f3d28607ddd7        160MB         45.3MB
ubuntu@ip-172-31-10-98:~$ docker rmi 247650787b97
Untagged: tomcat:latest
Deleted: sha256:247650787b97751d68aa2d532d6e09111c36946d9a50ea0e5071e19facc1e5e7
ubuntu@ip-172-31-10-98:~$ docker images

IMAGE                  ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest     0e760fdfbc48       25.9kB         9.49kB    U
jenkins/jenkins:lts    01c992ffef29        799MB          289MB
nginx:latest           5aca99593157        241MB           66MB    U
sonarqube:latest       35bedac3f40c       2.52GB         1.08GB
ubuntu:latest          f3d28607ddd7        160MB         45.3MB
ubuntu@ip-172-31-10-98:~$
```

6. Inspect the Jenkins image, SonarQube image.

Inspect Jenkins Image : docker inspect jenkins/jenkins:lts

```
ubuntu@ip-172-31-10-98:~$ docker inspect jenkins/jenkins:lts
[
    {
        "Id": "sha256:01c992ffef29dcf41c7164e8c16285c657c2368c4943dda8d68c93fdf54447d5",
        "RepoTags": [
            "jenkins/jenkins:lts"
        ],
        "RepoDigests": [
            "jenkins/jenkins@sha256:01c992ffef29dcf41c7164e8c16285c657c2368c4943dda8d68c93fdf54447d5"
        ],
        "Comment": "buildkit.dockerfile.v0",
        "Created": "2026-05-13T17:19:33.598838948Z",
        "Config": {
            "User": "jenkins",
            "ExposedPorts": {
                "50000/tcp": {},
                "8080/tcp": {}
            },
            "Env": [
                "PATH=/opt/java/openjdk/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
                "LANG=C.UTF-8",
                "JENKINS_HOME=/var/jenkins_home",
                "JENKINS_SLAVE_AGENT_PORT=50000",
                "REF=/usr/share/jenkins/ref",
```

Inspect SonarQube Image : docker inspect sonarqube

```
ubuntu@ip-172-31-10-98:~$ docker inspect sonarqube
[
    {
        "Id": "sha256:35bedac3f40cda75969890da59b17d577770844fe6ef659206c678a8e00921c7",
        "RepoTags": [
            "sonarqube:latest"
        ],
        "RepoDigests": [
            "sonarqube@sha256:35bedac3f40cda75969890da59b17d577770844fe6ef659206c678a8e00921c7"
        ],
        "Comment": "buildkit.dockerfile.v0",
        "Created": "2026-05-11T22:13:40.873390969Z",
        "Config": {
            "User": "sonarqube",
            "ExposedPorts": {
                "9000/tcp": {}
            },
            "Env": [
                "PATH=/opt/java/openjdk/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
                "JAVA_HOME=/opt/java/openjdk",
                "LANG=en_US.UTF-8",
                "LANGUAGE=en_US:en",
                "LC_ALL=en_US.UTF-8",
                "JAVA_VERSION=jdk-25.0.3+9",
```

i-08ba2672d4332489d (docker-test)

7. Run Jenkins container and run one sample job.

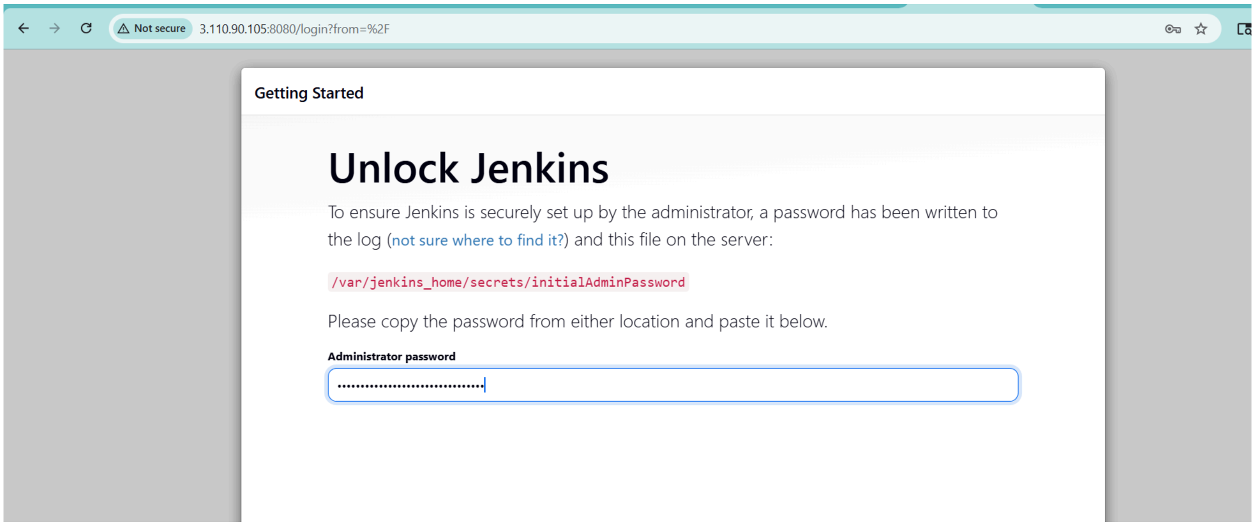
Run Jenkins Container

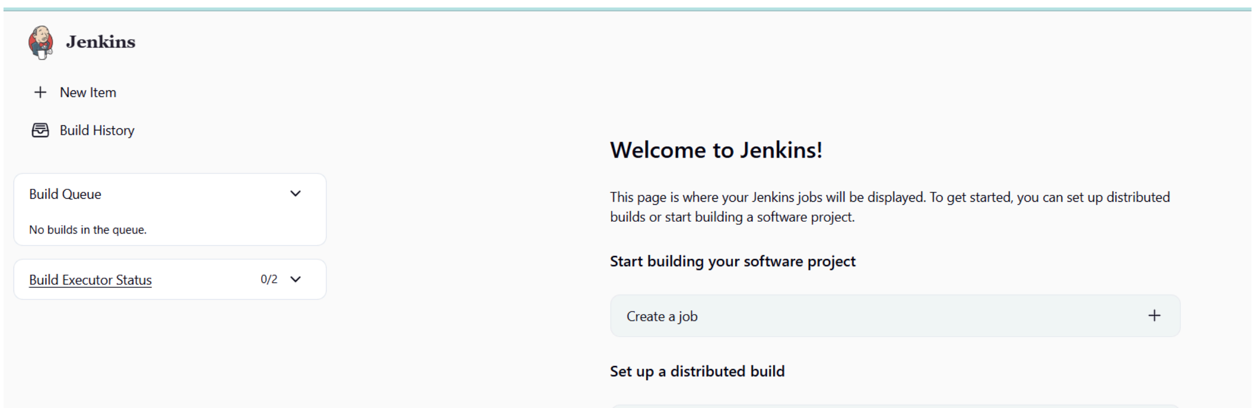
```
ubuntu@ip-172-31-10-98:~$ docker run -d \
--name jenkins \
-p 8080:8080 \
-p 50000:50000 \
jenkins/jenkins:lts
ec8124ebbbb394dfda8bdeeccd0c76fb278a61606de8ccc4d7af586cc316da19
ubuntu@ip-172-31-10-98:~$ docker ps
```

Check container:

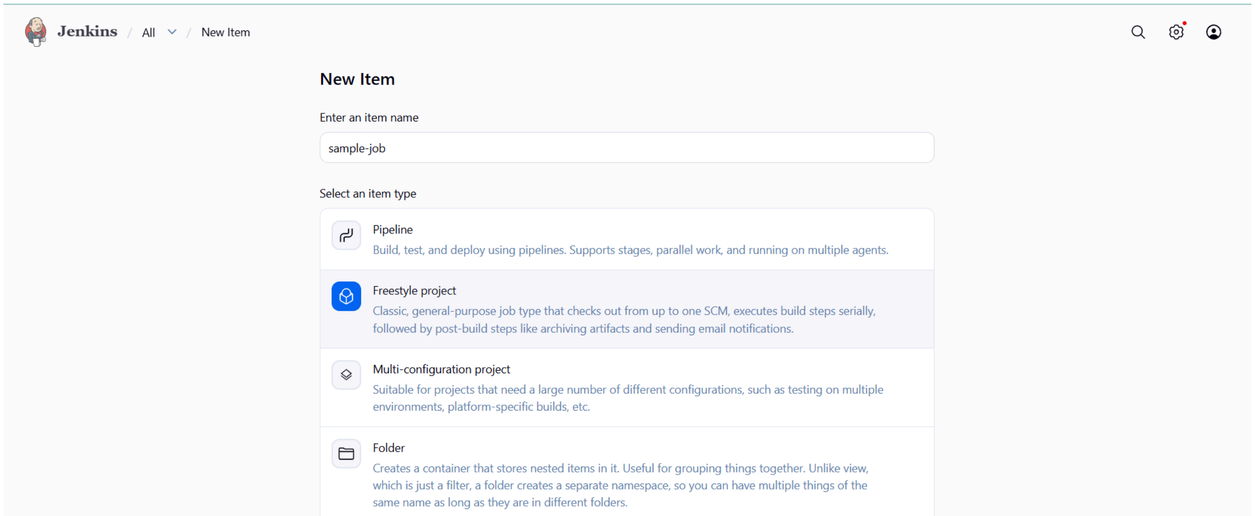
```
ubuntu@ip-172-31-10-98:~$ docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED          STATUS          PORTS
                 NAMES
ec8124ebbbb3   jenkins/jenkins:lts   "/usr/bin/tini -- /u…"   12 seconds ago   Up 12 seconds   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:
]:50000->50000/tcp   jenkins
9dddb9bd4569   nginx                 "/docker-entrypoint.…"   6 minutes ago    Up 6 minutes    0.0.0.0:81->80/tcp, [::]:81->80/tcp
                 nginx-container
ubuntu@ip-172-31-10-98:~$
```

docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword
Open Jenkins : ip:8080

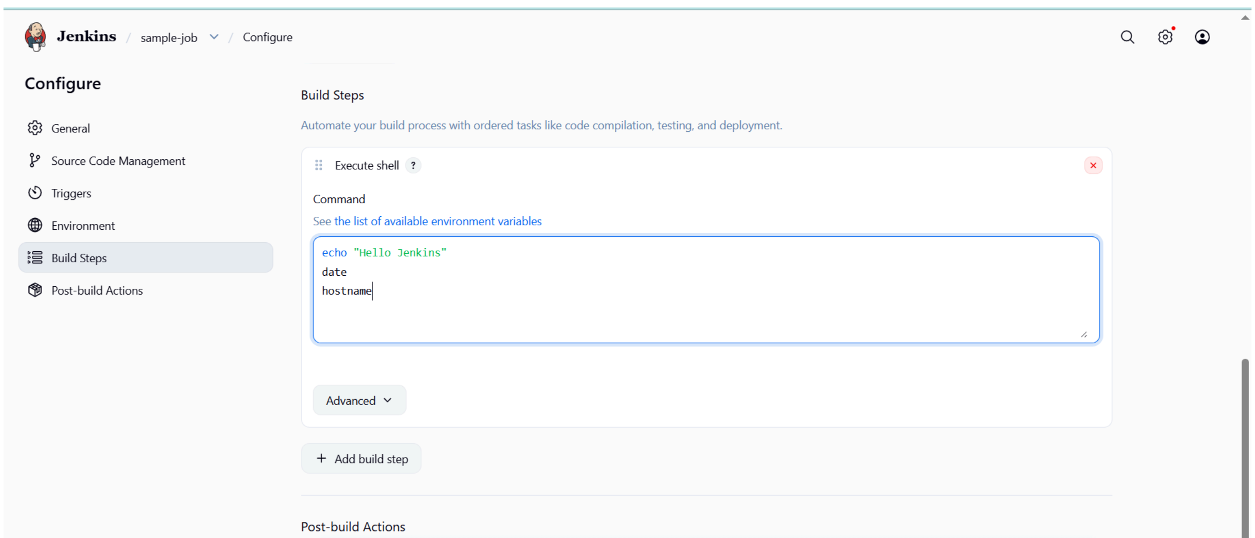




Create Sample Jenkins Job :



Build step:



Save & build now

Check the console output:


Jenkins
sample-job
#1
Status
Changes
Console Output
Edit Build Information
Delete build '#1'
Timings
Console
Download
Copy
View as plain text
Started by user bhargav radharapu
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/sample-job
[sample-job] $ /bin/sh -xe /tmp/jenkins10964721977034230961.sh
+ echo Hello Jenkins
Hello Jenkins
+ date
Sat May 30 18:22:59 UTC 2026
+ hostname
ec8124ebbbb3
Finished: SUCCESS